

PRIMEASIA UNIVERSITY

Department of Computer Science & Engineering

PROJECT PROPOSAL

SMART ATTENDANCE MANAGEMENT SYSTEM

Course Code: CSE-232

Course Title: Database System Design (Lab)

Submitted To:

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1. Title

Smart Attendance Management System Using Web Technology

2. Introduction

The Smart Attendance Management System is a modern web-based solution designed to automate and simplify the traditional attendance process. In many educational institutions, attendance is still recorded manually, which consumes time and increases the possibility of human errors. This system eliminates these problems by using digital technology to store and manage attendance efficiently.

The system is developed using PHP and MySQL, which ensures fast processing, secure data storage, and easy accessibility. It provides a simple interface where users can mark attendance, generate reports, and analyze student performance. Overall, the system aims to improve efficiency, accuracy, and data management in educational environments.

3. Objectives

- To automate the attendance recording process
- To minimize human errors
- To store and manage student data efficiently
- To generate attendance reports
- To calculate attendance percentage
- To provide a user-friendly system

4. Problem Statement

Traditional attendance systems rely heavily on manual processes, which are inefficient and prone to errors. Teachers often face difficulties in maintaining records, calculating attendance percentages, and generating reports. Additionally, paper-based systems are not secure and can be easily lost or damaged.

This project aims to solve these problems by providing a digital system that ensures accuracy, efficiency, and security. The system reduces manual workload and allows quick access to attendance records.

5. Methodology

The system is developed following a structured approach. First, the requirements are analyzed to understand user needs. Then, a proper database and system design is created. After that, the system is implemented using HTML, CSS, PHP, and MySQL.

Once development is complete, the system is tested to identify and fix errors. Finally, the system is deployed using XAMPP, allowing users to interact with it through a web browser. This step-by-step approach ensures the system is reliable and efficient.

6. Database Design

The system uses a relational database structure consisting of three main tables:

- Students Table → stores student information
- Attendance Table → stores daily attendance records
- Admin Table → manages login credentials

Relationships are maintained using keys to ensure data integrity. This design allows efficient storage, retrieval, and management of data.

7. Requirements

Software Requirements:

- XAMPP
- Web Browser
- Visual Studio Code

Hardware Requirements:

- Computer/Laptop
- Minimum 4GB RAM

8. Expected Outcome

The system will provide a complete digital solution for attendance management. Users will be able to mark attendance quickly, generate reports, and calculate attendance percentages automatically.

This system will reduce manual workload, improve accuracy, and ensure proper data management.

9. Timeline

Requirement Analysis – 1 Day

Database Design – 1 Day

Development – 3 Days

Testing – 1 Day

Final Implementation – 1 Day

10. Conclusion

The Smart Attendance Management System is an efficient and reliable solution for managing attendance in educational institutions. It reduces errors, saves time, and improves data organization.

In the future, this system can be enhanced by integrating advanced technologies such as biometric authentication or face recognition, making it more intelligent and automated.